(iv) Intercepts form:

If the intercepts of a line on the$x$-axis and the y-axis are known then the equation of the line can also be found using intercepts. Suppose x-intercept $OA = a$ and y-intercept $OB = b$, where a and b are non-zero, then the line passes through two points $A(a, 0)$ and $B(0, b)$ is

$$\frac{y-0}{b-0} = \frac{x-a}{0-a}$$

$$\frac{x}{a} + \frac{y}{b} = 1 \qquad (6.8)$$

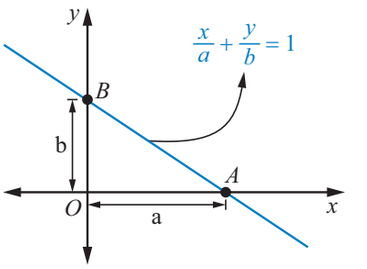


Figure 6.21

The above equation is called an intercept form.

Lines that pass through the origin or which are horizontal or vertical or violate the nonzero condition on a or b cannot be represented in this form.

In most of the cases this form is used to draw the graph of the line in easy way.

(v) Normal form:

Let A and B be the intercepts made by the line.

Let p be the length of the normal OP drawn from the origin to a line AB, which makes an angle α with the x-axis.

$$\text{In right } \Delta OPA, \frac{OP}{OA} = \cos\alpha \text{ and}$$

$$\text{in right } \Delta OPB, \frac{OP}{OB} = \cos\left(\frac{\pi}{2} - \alpha\right) = \sin\alpha$$

$$\Rightarrow \frac{1}{OA} = \frac{\cos\alpha}{p}$$

$$\text{and } \frac{1}{OB} = \frac{\sin\alpha}{p}$$

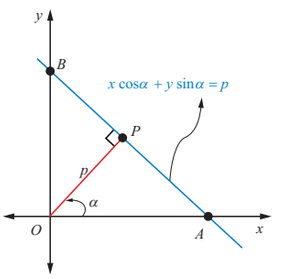


Figure 6.22

Using the above data in intercepts form

$$\frac{x}{OA} + \frac{y}{OB} = 1$$

$$\text{We get, } \frac{x\cos\alpha}{p} + \frac{y\sin\alpha}{p} = 1$$

$$\Rightarrow x\cos\alpha + y\sin\alpha = p \qquad (6.9)$$

is called the normal form of equation.

If p is positive in all positions of the line and if α is always measured from x-*axis* in the positive direction, this equation holds in every case as shown in the figure.

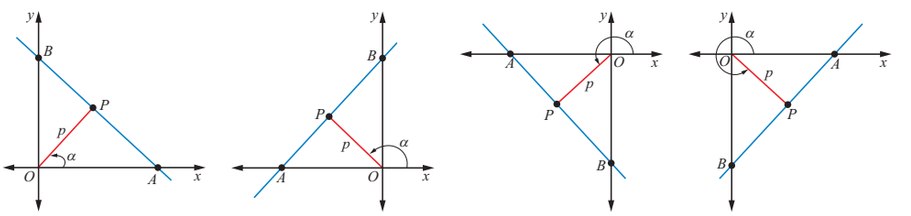


Figure 6.23